Continuity & Differentiability: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Continuity at a point

Formulas

$$\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) = f(a)$$

KEY POINTS

- Graph: drawn through $x = a$ without lifting the pen
- a must be in the domain; continuity is discussed only in the domain
- End point of domain: only the inside one-sided limit is compared with $f(a)$
- Left continuous: $f(a^-) = f(a)$; right continuous: $f(a^+) = f(a)$
- Continuous $\Rightarrow$ limit exists; converse false

STEPS

1. Find $f(a)$ from the defining piece
2. Find LHL and RHL from the pieces on each side
3. Equate all three; solve for the constants

2. Algebra of continuous functions

KEY POINTS

- Polynomial, trigonometric, exponential, logarithmic functions: continuous in their domains
- f, g continuous $\Rightarrow f \pm g,\ fg,\ f/g\ (g(a) \neq 0)$ continuous
- f continuous, g discontinuous $\Rightarrow f \pm g$ discontinuous; fg, f/g: check
- Both discontinuous: nothing definite (e.g. $[x] + \{x\} = x$)

Watch out

$x[x]$ is continuous at 0 (zero factor) but not at 1

3. Continuity in an interval and checkpoints

KEY POINTS

- (a, b): continuous at every point
- $[a, b]$: continuous in (a, b), right continuous at a, left continuous at b

STEPS

1. Points where the definition changes
2. $[f(x)]$, $\{f(x)\}$: where $f(x) \in I$
3. $\operatorname{sgn} f(x)$: where $f(x) = 0$
4. Zeros of denominators, breaks of domain

Watch out

At a checkpoint the function may still be continuous: always verify
For $\{f\}$ and $[f]$ count only the points inside the given interval

4. Special functions: Q/Q', $[f(x)]$, built-in limit

KEY POINTS

- g on Q, h on Q': continuous only where $g(x) = h(x)$
- $[f(x)]$ by graph: break where the graph crosses $y = n$ or touches it from below
- Touching $y = n$ from above (local min at n): $[f]$ stays continuous
- Built-in limit: split $|x| < 1$, $|x| = 1$, $|x| > 1$; then check $x = \pm 1$

Formulas

$$\lim_{n\to\infty} x^{2n} = 0\ (|x|<1),\ 1\ (|x|=1),\ \infty\ (|x|>1)$$

5. Types of discontinuity

KEY POINTS

- Removable: limit exists; missing point ($f(a)$ undefined) or isolated point ($f(a) \neq$ limit)
- Non-removable: limit does not exist
- Finite (jump): both one-sided limits finite, unequal; jump $= |\text{RHL} - \text{LHL}|$
- Infinite: a one-sided limit is $\pm\infty$; oscillatory: $\sin\frac{1}{x}$ at 0

Watch out

Removable discontinuity is fixed by redefining $f(a) =$ limit

6. Composite function $f(g(x))$

STEPS

1. Points where g is discontinuous
2. Points x where $g(x)$ equals a discontinuity point of f
3. Or write $f(g(x))$ piecewise and check the junctions

Watch out

Points outside the domain of $f(g(x))$ (e.g. $x = 1$ for $f(f(x))$, $f = \frac{1}{1-x}$) count as discontinuities

7. Extreme value theorem and IMVT

KEY POINTS

- EVT: continuous on $[a, b] \Rightarrow$ attains min m and max M
- IMVT: continuous on $[a, b]$ takes every value between $f(a)$ and $f(b)$
- $f(a)f(b) < 0 \Rightarrow$ at least one root in (a, b) (sufficient, not necessary)

STEPS

1. Write $g(x) =$ LHS $-$ RHS (or $f(x) - x$, $f(x + k) - f(x)$)
2. Check continuity of g on the closed interval
3. Show $g(a)g(b) \leq 0$

Watch out

Continuous function taking only integer (or rational) values is constant

8. Derivability at a point

Formulas

$$\text{RHD} = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}, \qquad \text{LHD} = \lim_{h\to 0^+} \frac{f(a-h)-f(a)}{-h}$$

KEY POINTS

- Derivable: LHD $=$ RHD $=$ finite; unique tangent of finite slope
- Also $f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$; $\lim_{h\to 0} \frac{f(a+ph)-f(a+qh)}{h} = (p-q)f'(a)$
- End point: only the inside one-sided derivative
- Derivable $\Rightarrow$ continuous; discontinuous $\Rightarrow$ not derivable

Watch out

Use $f(a)$, not the limit, in the difference quotient

9. LHD, RHD and the graph

KEY POINTS

- Equal and finite: derivable and continuous
- Finite, unequal: continuous, not derivable (sharp corner)
- One infinite: not derivable; continuity must be checked separately
- $x^{1/3}$: vertical tangent, no corner; $|x|^{1/3}$: corner

10. Algebra of derivable functions

KEY POINTS

- Both derivable: $f \pm g,\ fg,\ f/g\ (g(a) \neq 0)$ derivable
- f derivable, g not: $f \pm g$ not derivable; fg check
- Both non-derivable: nothing definite
- $f(a) = 0$, f derivable, g continuous at $a \Rightarrow fg$ derivable at a

11. Derivability in an interval and checkpoints

KEY POINTS

- $[a, b]$: derivable in (a, b), RHD at a and LHD at b finite

STEPS

1. All continuity checkpoints (discontinuous $\Rightarrow$ not derivable)
2. $|f(x)|$: simple zeros of f
3. Vertical tangents, e.g. $(x - a)^{1/3}$
4. max/min of functions: intersection points (draw the graph, count corners)

Watch out

Repeated zero: $|x - a|^2$, $|x - a|(x - a)$ are derivable at a

12. $g(x)|f(x)|$, Q/Q' and $x^p \sin \frac{1}{x}$

KEY POINTS

- $g(x)|f(x)|$, $f(a) = 0$: derivable at a iff $g(a) = 0$
- Q/Q' function: derivable only where $g = h$ and $g' = h'$
- $x^p \sin \frac{1}{x}$: continuous $p > 0$, derivable $p > 1$, f' continuous $p > 2$

13. Quick concept and non-uniform functions

Formulas

$$\text{LHD} = \lim_{x \to a^-} f'(x), \qquad \text{RHD} = \lim_{x \to a^+} f'(x)$$

STEPS

1. Check continuity at the junction first
2. Differentiate each piece on its open interval
3. Compare the one-sided values of f'

Watch out

Not valid if discontinuous or if $\lim f'$ oscillates ($x^2 \sin \frac{1}{x}$)
Composite: write $f(g(x))$ piecewise first, then check

14. max / min over $[a, x]$

STEPS

1. Sketch $f(t)$ from $t = a$
2. Max: $y = f(x)$ while f is at a new high, else the highest value so far
3. Min: $y = f(x)$ while f is at a new low, else the lowest value so far
4. Check continuity and LHD, RHD at the junctions

15. Functional equations

Formulas

$$f(x+y)=f(x)+f(y)\Rightarrow f=kx;\quad f(x+y)=f(x)f(y)\Rightarrow f=e^{kx};\quad f(xy)=f(x)+f(y)\Rightarrow f=k\ln x$$

STEPS

1. Put $x=y=0$ (or 1) to find $f(0)$ (or $f(1)$)
2. $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$, apply the rule, get $f'(x)$ in terms of $f'(0)$
3. Or differentiate partially and put the other variable $=0$ or 1
4. Integrate and use $f(0)$ / $f(1)$

Watch out

$|f(x)-f(y)|\le(x-y)^2\Rightarrow f'(x)=0\Rightarrow f$ constant

16. Common mistakes checklist

STEPS

1. Did I check that the point is in the domain?
2. Did I compare with $f(a)$, not only LHL = RHL?
3. Did I check continuity before differentiability?
4. Did I use the quick concept only for a continuous function?
5. Did I count points only inside the given (open/closed) interval?
6. Did I check zeros of $|f|$ for repeated roots or a zero factor outside?